Tut 2

Ex. 8.2

$\boxed{\text{Tabular integration}}$

Q20 $\displaystyle\int \underline{t^2} e^{4t}\, dt$

+	t^2	$\underline{e^{4t}}$
−	$2t$	$\frac{1}{4}e^{4t}$
+	2	$\frac{1}{16}e^{4t}$
−	0	$\frac{1}{64}e^{4t}$

$$= t^2 \cdot \frac{1}{4}e^{4t} - 2t \cdot \frac{1}{16}e^{4t} + 2 \cdot \frac{1}{64}e^{4t} + C$$

$$= \frac{1}{4}t^2e^{4t} - \frac{1}{8}te^{4t} + \frac{1}{32}e^{4t} + C$$

R22

$$\int \underbrace{e^{-y}}_{\text{int.}} \underbrace{\cos y}_{\text{diff.}} \, dy$$

$+$	$\cos y$	e^{-y}
$-$	$-\sin y$	$-e^{-y}$
$+$	$-\cos y$	e^{-y}

$\int \square \, dx$

$$= -e^{-y}\cos y - (-\sin y \, e^{-y}) + \int -\cos y \cdot e^{-y} \, dy$$

$$I := \int e^{-y}\cos y \, dy$$

$$\Rightarrow I = -e^{-y}\cos y + e^{-y}\sin y - I$$

$$2I = -e^{-y}\cos y + e^{-y}\sin y$$

$$\therefore I = \frac{1}{2}\left(-e^{-y}\cos y + e^{-y}\sin y\right) + C$$

Q42 $\int \sin 2x \cos 4x \, dx$

① Use Product to Sum formula

$$\sin A \cos B = \frac{1}{2}\sin(A-B) + \frac{1}{2}\sin(A+B)$$

② Int. by part

	diff	Int.
+	$\sin 2x$	$\cos 4x$
−	$2\cos 2x$	$\frac{1}{4}\sin 4x$
+	$-4\sin 2x$	$-\frac{1}{16}\cos 4x$

$$\text{Integral} = \frac{1}{4}\sin 2x \sin 4x - 2\cos 2x\left(-\frac{1}{16}\cos 4x\right) + \int \frac{1}{4}\sin 2x \cos 4x \, dx$$

$$I := \int \sin 2x \cos 4x \, dx$$

$$I = \frac{1}{4}\sin 2x \sin 4x + \frac{1}{8}\cos 2x \cos 4x + \frac{1}{4} I$$

$$\frac{3}{4} I = \cdots$$

$$I = \cdots$$

<u>Q54</u> $\displaystyle\int \frac{xe^x}{(\underbrace{x+1}_{u})^2}\,dx$ $\qquad \dfrac{d}{dx}\left(\dfrac{x}{(x+1)^2}\right) = ??$

$$= \int \frac{(u-1)e^{u-1}}{u^2}\,du$$

$u = x+1$

$x = u-1$

$du = dx$

$$= \int \underbrace{u^{-1}}_{\text{diff.}} \underbrace{e^{u-1}}_{\text{int.}}\,du - \int u^{-2}e^{u-1}\,du$$

	diff.	int.
+	u^{-1}	e^{u-1}
−	$-u^{-2}$	e^{u-1}

$$= u^{-1}e^{u-1} + \int u^{-2}e^{u-1}\,du - \int u^{-2}e^{u-1}\,du$$

$$= \frac{e^{u-1}}{u} + C = \frac{e^x}{x+1} + C$$

Ex. 8.3

Q40 $\int e^x \sec^3(e^x)\,dx$

$= \int \sec^3 u\,du$ $\qquad u = e^x$, $du = e^x\,dx$

$= \int \sec u \cdot \sec^2 u\,du$ $\qquad \int \sec^2 u\,du = \tan u$

$= \tan u \sec u - \int \tan u \sec u \tan u\,du$

$= \tan u \sec u - \int \tan^2 u \sec u\,du$

$\boxed{\sec^2 u - 1}$

$\downarrow$

$(\sec^2 u - 1)\sec u$

$\downarrow$

$\sec^3 u - \sec u$

$= \tan u \sec u - \int \sec^3 u\,du + \int \sec u\,du$

$I := \int \sec^3 u\,du$ $\qquad$ ssc tsc

$\Rightarrow I = \tan u \sec u - I + \ln|\sec u + \tan u|$

$2I = \cdots$

$I = \cdots + C$ $\qquad$ # Try Q42

Q59 $\int \sin^2\theta \cos 3\theta \, d\theta$

$$= \int \left(\frac{1-\cos 2\theta}{2}\right)\cos 3\theta \, d\theta$$

$$= \frac{1}{2}\int \underbrace{\cos 3\theta \, d\theta}_{\text{easy formula}} - \frac{1}{2}\int \underbrace{\cos 2\theta \cos 3\theta}_{\text{Product to Sum}} \, d\theta$$

$\cos A \cos B$

$$= \frac{1}{2}\left(\cos(A-B) + \cos(A+B)\right)$$

Q60 $\int \cos^2 2\theta \sin\theta \, d\theta$

~~① Half-Angle Formula~~ ① $\boxed{\cos 2\theta = 2\cos^2\theta - 1}$

~~② Half-Angle~~ ② $\cos^2 2\theta = \frac{\cos 4\theta + 1}{2}$

Q61 $\int \cos^3\theta \underbrace{\sin 2\theta}_{2\sin\theta\cos\theta} \, d\theta$

$$= 2\int \sin\theta \cos^4\theta \, d\theta$$

Q63 $\int \sin\theta \cos\theta \cos 3\theta \, d\theta$

Product to Sum

$$\sin\theta\cos\theta = \frac{1}{2}\left(\sin(\theta-\theta) + \sin(\theta+\theta)\right)$$

$$= \frac{1}{2}\sin 2\theta$$

$$\frac{1}{2}\int \sin 2\theta \cos 3\theta \, d\theta$$

Product to Sum

Ex. 8.4

$x^2+1 \rightarrow x = \tan\theta$

$\sqrt{x^2-1} \rightarrow x = \sec\theta$

$\sqrt{1-x^2} \rightarrow x = \sin\theta$

Q49 - Q52

Q48 $\int \frac{\sqrt{x-2}}{\sqrt{x-1}} \, dx = \int \frac{\sqrt{u-1}}{\sqrt{u}} \, du$ $\quad u = x-1$

$t = \sqrt{u}$

$dt = \frac{1}{2\sqrt{u}} \, du$

$$= 2\int \sqrt{t^2-1} \, dt$$